**Title:** An open-source, provenance-aware framework for reconstructing missing CT dose indices from DICOM metadata

**Running title:** Provenance-aware CT dose reconstruction

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**Keywords:** computed tomography; CTDIvol; size-specific dose estimate; radiation dose structured report; open data; reproducibility

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**Conflicts of interest.** The author is the representative of LISIT Co., Ltd. (Tokyo, Japan) and Chief Executive Officer of TexelCraft OÜ, and this work was conducted under the supervision of LISIT Co., Ltd. The software is released under the MIT License and the author has a commercial interest in downstream products that may use it. No patient or customer data were used.

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**Ethics approval and consent.** Not applicable. This study used only publicly available, de-identified

imaging data (TCIA collections released under Creative Commons licenses) and third-party sample files. No institutional review board approval and no informed consent were required.

**Data and code availability.** The software is available at https://github.com/Institute-of-One/ctdose-core under the MIT License. The version used in this study (version 0.2.0) is archived at Zenodo (https://doi.org/10.5281/zenodo.21636719).

**Supplementary Material.** One supplementary file accompanies this Technical Note, containing the verification of the size layer and of the RDSR parser, the auditable diagnostic-reference-level comparison, the sample licensing and provenance, the scanner models the coefficient table does not cover, the stratified accuracy detail, the size-correction and effective-dose results, the data-quality findings, and the GE validation arm in full. It contains no figures.

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**Author contributions.** S.Y.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing — original draft, writing — review and editing, visualization.